THE WAGNER GRAPH REFUTES AN EL ZEIN–MORTADA LOCAL-GIRTH CONJECTURE

Abstract. El Zein and Mortada list as conjectured that every $(3,3)$-saturated subcubic graph G with $g_3(G) = 4$ is $(1,2,2,2,2)$-packing colorable. The Wagner graph is a counterexample. Every vertex has local girth 4, and the graph has diameter 2 and independence number 3; hence a $(1,2,2,2,2)$-packing coloring can cover at most $3+4=7$ of its eight vertices. We also prove that every subcubic graph on at most seven vertices is $(1,2,2,2,2)$-packing colorable, so order eight is minimum. Earlier computation of Gastineau and Togni already implied the existence of such an obstruction. The present contribution is a standard explicit witness and an elementary proof of minimum order.

Throughout, graphs are finite and simple, and vertices in different components are taken to be at infinite distance. For a nondecreasing sequence $S = (s_1, \ldots, s_k)$, an *S-packing coloring* of a graph G is a partition

$$V(G) = V_1 \,\dot\cup\, \cdots \,\dot\cup\, V_k$$

such that $\operatorname{dist}_G(u,v) > s_i$ for all distinct $u, v \in V_i$. The local girth $g(v)$ is the length of a shortest cycle containing v, with $g(v) = \infty$ if no such cycle exists, and

$$g_3(G) = \max\{g(v) : d_G(v) = 3\}.$$

Following El Zein and Mortada, a degree-three vertex is *heavy* if all its neighbors have degree three, and a subcubic graph is $(3,k)$-saturated if every heavy vertex has at most k heavy neighbors. Thus every subcubic graph is $(3,3)$-saturated.

In version 1 of [2], Table 1 lists the following assertion in its conjectured column; it is the $g_3(G) = 4$ case of the first question in [1, Problem 1]:

> *Every $(3,3)$-saturated subcubic graph G with $g_3(G) = 4$ is $(1,2,2,2,2)$-packing colorable.*

We refer to this assertion as the El Zein–Mortada local-girth conjecture.

Let W be the Wagner graph, with vertex set $\mathbb{Z}_8$ and edge set

$$E(W) = \big\{\{i, i+1\} : i \in \mathbb{Z}_8\big\} \cup \big\{\{i, i+4\} : i = 0,1,2,3\big\},$$

where all additions are modulo 8.

Theorem 1. *The graph W is a $(3,3)$-saturated cubic graph with $g_3(W) = 4$ that is not $(1,2,2,2,2)$-packing colorable. Moreover, every subcubic graph of order at most seven is $(1,2,2,2,2)$-packing colorable. Consequently, W is a*

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minimum-order counterexample to the El Zein–Mortada local-girth conjecture and to the first question in [1, Problem 1].

Proof. The graph W is cubic, hence $(3,3)$-saturated. For every $i \in \mathbb{Z}_8$,

$$i,\ i+1,\ i+5,\ i+4$$

induces a 4-cycle. The neighbors $i-1, i+1, i+4$ of i are pairwise nonadjacent, so no triangle contains i. Thus every vertex has local girth exactly 4, and $g_3(W) = 4$.

Every two vertices are at distance at most two. Cyclic differences 1 and 4 give edges, difference 2 is realized along the 8-cycle, and for difference 3 one may use the path $i, i+4, i+3$. Since 0 and 2 are nonadjacent, $\operatorname{diam}(W) = 2$.

The set $\{0,2,5\}$ is independent. An independent set of order four in the spanning cycle C_8 must be one of

$$\{0,2,4,6\}, \qquad \{1,3,5,7\},$$

and each contains antipodal edges of W. Hence $\alpha(W) = 3$. In a $(1,2,2,2,2)$-packing coloring, the color-1 class therefore has at most three vertices. Each of the four color-2 classes has at most one vertex because $\operatorname{diam}(W) = 2$. Such a coloring covers at most $3 + 4 = 7 < 8$ vertices, a contradiction.

It remains to prove minimum order. Colors may be reused in different components, so it suffices to color each connected component. A component of order at most four can be colored by assigning its vertices distinct color-2 classes. Let G be connected with $5 \le |V(G)| \le 7$. If $\Delta(G) \le 2$, then $\chi(G) \le 3$. If $\Delta(G) = 3$, Brooks' theorem also gives $\chi(G) \le 3$, since G is neither K_4 nor an odd cycle. A largest class in a proper 3-coloring has at least $\lceil |V(G)|/3 \rceil$ vertices. Assign that independent class color 1 and assign the remaining vertices distinct color-2 classes. This is possible because

$$|V(G)| - \left\lceil \frac{|V(G)|}{3} \right\rceil \le 4 \qquad (5 \le |V(G)| \le 7).$$

Thus every subcubic graph of order at most seven is $(1,2,2,2,2)$-packing colorable. □

Priority and scope

Gastineau and Togni's exhaustive table records two cubic graphs of order 8 that are $(1,2,2,2,2,2)$-packing colorable but not $(1,2,2,2,2)$-packing colorable [3, Table 1]. The Wagner graph is one of them: the classes

$$\{0,2,5\},\ \{1\},\ \{3\},\ \{4\},\ \{6\},\ \{7\}$$

give a $(1,2,2,2,2,2)$-packing coloring, while Theorem 1 excludes $(1,2,2,2,2)$.

Their computation already implied a negative answer to the later local-girth question. Indeed, if a vertex of an order-eight cubic graph belonged to no triangle or 4-cycle, its radius-two ball would contain

$$1 + 3 + 3 \cdot 2 = 10$$

distinct vertices. The absence of triangles makes the first and second layers disjoint, while the absence of 4-cycles makes the six vertices in the second layer distinct across the three branches. Thus every vertex of an order-eight cubic graph has local girth at most 4.

Only the $(1,2,2,2,2)$ assertion is refuted. The Wagner graph is $(1,1,2,3)$-packing colorable, for example with color classes

$$\{0,2,5\}, \qquad \{1,3,6\}, \qquad \{4\}, \qquad \{7\}.$$

It therefore does not address the second question in [1, Problem 1] or Conjecture 1 of that paper.

References

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[3] N. Gastineau and O. Togni, *S-packing colorings of cubic graphs*, Discrete Math. **339**(10) (2016), 2461–2470, doi:10.1016/j.disc.2016.04.017.